into Laplace space where a differential equation becomes a linear equation. Thus in Laplace space equation 17.18 becomes:

$$sN_{21} - \widehat{L}N_{21}^{0} = \lambda_{11}\widehat{L}N_{11} - (\varphi\sigma_{21} + \lambda_{21})\,\widehat{L}N_{21} \tag{17.45}$$

The system of now linear equations is solved with respect to the desired nuclide and the solution is transferred back into normal space if the inverse transform is known.

17.4. TARGET CONSIDERATIONS

The success of an irradiation experiment depends to a large extent on the target considerations. In many nuclear experiments as much time has to be devoted to the target preparation as to the rest of the experiment. If the purpose is to produce a radionuclide for a simple tracer experiment, the consideration in this section may be sufficient. If, however, the requirements are a product of extreme purity, very high specific activity, or very short half-life, special techniques must be used, which are discussed in §§17.5–17.7. The production of atoms of the heavy elements also require fast and efficient production and separation methods.

17.4.1. Physical properties

In thin targets the flux and energy of the projectile do not change during passage through the target and calculation of e.g. production rates are straight forward by applying e.g. equation 17.14 using k from 17.12. If, however the projectile energy is considerably changed (in a thick target) it is more complicated to calculate the yield as the variation of energy with distance in the target and the variation of cross section with energy must be taken into account.

The exact specification of when a target is thin depends on the experiment; in most cases thin means a target with a surface weight of not more than a few mg/cm^2 or a thickness of a few μm or less. The *surface weight* (or *linear density*) is $x\rho$, where x is the target thickness and ρ its density. Thin targets are usually used in accelerator irradiations and are made of a solid material, either metal or compound. In order to have sufficient mechanical strength the target is usually supported on a *backing material* such as aluminum. The backing material may be part of a solid target holder which has provisions for cooling to dissipate the heat developed in the bombardment. Thin targets are fixed on the backing material through electrolytic precipitation, vacuum deposition, mass spectrometric collection (for isotopically pure targets), or a number of other techniques.

In reactor irradiations it is more common to use thicker targets in order to obtain higher yields. In traversing the target the effective flux (i.e. $\varphi\sigma$) does not decrease as rapidly for neutrons as for charged particles and, therefore, thicker targets can still be considered "thin" in neutron irradiations. However, if the product σN is large, the neutron flux may decrease appreciably at the center of the target, especially at the neutron energies where cross sections are large.

The target material can be a solid, liquid, or gas. For reactor irradiations it is common to place the target material in a container of polypropylene or of some metal of relatively low neutron capture cross-section. Some other materials should be avoided since they contain elements wich will themselves be activated in the neutron flux and thus complicating the handling of the sample. Such examples are pyrex and soda glass because of the reactions ^{10}B (n,α) ^{7}Li (which has a large σ_{capt}) and ^{23}Na (n,γ) ^{24}Na (which yields a long-lived γ-product). In silica some γ-active ^{31}Si ($t_{½}$ 2.62 h) is produced ($\sigma_{capt} = 0.11$ b for thermal n). When metal sample holders, usually aluminum and magnesium are used, some γ-activity is produced; however, the cross-sections for the formation of the products ^{28}Al and ^{27}Mg are small, and these nuclides decay rapidly ($t_{½} = 2.24$ and 9.46 min, respectively). The primary requirement for the container is that it should be leak-tight and dissipate heat energy at a rate sufficient to avoid melting.

Another factor that affects the yields of an irradiation is the extent of *burn-out* of the target by the irradiation. If the number of target atoms at the beginning of the irradiation is N^0_{target} and at time t_{irr} is N_{target}, then the burn-out can be estimated with the relation

$$N_{target} = N^0_{target} (1 - \sigma_{tot}\, \varphi\, t_{irr}) \tag{17.46}$$

where σ_{tot} includes all processes whereby the target atoms are reduced. Burn-out can be neglected when $\sigma_{tot}\, \varphi\, t_{irr} \ll 1$.

In the example of the irradiation of the 0.3 mm thick gold foil in §11.1, from (11.2) we can calculate that the flux reduction is 17%; for a 0.01 mm thick foil it would be 0.62%, but for a 1 mm foil, 44%. The target thickness does not enter into the burn-out, eqn. (17.46). For a 10 min irradiation of 10^{13} n cm^{-2} s^{-1}, the burn-out is negligible. However, a one-month irradiation at the same flux yields 0.26% burn-out, while a month at a flux of 10^{15} gives about 20% burn-out.

Finally, since the production cross sections of the heaviest elements are low, usually in the pb − nb range, synthesis of these require that an actinide target of up to 1 mg/cm^2 thickness on a 1−5 μm backing is irradiated with a heavy ion beam of 1−5 particle μA. Although only a minor part of the beam energy is dissipated when it passes through the target a lot of heat is released. A recent development is therefore rotating target wheels with a number of (usually three) banana shaped targets. When the wheel rotates one target on the wheel is hit by the particle beam while the others are cooled. The recoiling reaction products are then usually swept away from the backside of the targets by a gas-jet transport system.

17.4.2. Chemical properties

There are two ways to produce a pure radionuclide not contaminated with any other radioactivity. An extremely pure target can be used with a reaction path which is unique. Alternatively, the radioactive products can be purified after the end of the bombardment.

For example, a 10 g sample of zinc irradiated for one week with 10^{13} n cm^{-2} s^{-1} yields a sample of ^{65}Zn ($t_{½}$ 244 d) with 7.1×10^9 Bq. If, however, the zinc target is contaminated with 0.1% of copper, in addition to the zinc activity, 3.0×10^9 Bq of ^{64}Cu ($t_{½}$ 12.7 h) is formed. In another example element $_{102}$No was believed to be discovered initially in a bombardment of a target of curium by carbon ions. The observed activity, however, was later found to be due to products formed due to the small amount of lead impurity in the target. Similarly, in neutron activation of samarium the target must be very free of europium contamination because of the larger europium reaction cross-sections. Handbooks of activation analysis often contain information on the formation of interfering activities from impurities.

Even a chemically pure target may yield products of several elements, particularly in cyclotron irradiation, where many reaction paths are often possible. In the bombardment of magnesium with deuterons, the following reactions occur:

$$^{24}\mathrm{Mg}(79\%)(\mathrm{d},\alpha)^{22}\mathrm{Na}(t_{½}\ 2.6\ \mathrm{y}) \tag{17.47}$$

$$^{25}\mathrm{Mg}(10\%)(\mathrm{d},\mathrm{n})^{26}\mathrm{Al}\left(t_{½}\ 7.2 \times 10^5\mathrm{y}\right) \tag{17.48}$$

$$^{26}\mathrm{Mg}(11\%)(\mathrm{d},\alpha)^{24}\mathrm{Na}(t_{½}\ 14.66\ \mathrm{h}) \tag{17.49}$$

A short time after the irradiation ^{24}Na is the predominant activity; after longer cooling times the ^{22}Na is always contaminated by ^{26}Al. In such a case chemical purification after the irradiation is required to yield a pure product.

A suitable target material may be unavailable, or very expensive because of a low natural abundance. ^{45}Ca, which is the most useful calcium radioisotope ($t_{½}$ 164 d), is produced through n,γ capture in ^{44}Ca; unfortunately, ^{44}Ca only occurs to 2.1% in nature. While irradiation of natural calcium may yield a specific activity of $\sim$0.3 TBq g^{-1}, isotope enrichment before irradiation to essentially pure ^{44}Ca yields $\leq$20 TBq g^{-1}.

When the target is a chemical compound its radiation stability becomes important. In general organic compounds are not stable towards either neutron or charged particle bombardment. However, with neutrons a very particular and useful kind of reaction, known as the Szilard-Chalmers reaction (see §17.6.3), may take place. Simple inorganic compounds like NaCl are stable, and for some inorganic compounds like Na_2CO_3 the radiation decomposition (to Na_2O and CO_2) are of little importance. However, it may be noted that irradiations producing e.g. alpha active nuclides may eventually cause the inner helium pressure increase so much that crystals etc may rupture.

When chemistry experiments will be performed on the products from the nuclear reaction the products have to be transferred fast and continuously to the chemical separation system. As mentioned above, this is usually done by sweeping the back side of

the target with a gas (usually He) containing aerosol particles (often KCl). The formed products have enough energy to recoil out of the target, if it is thin enough, and recoiling ionized atoms attach them selves to the aerosol particles. They can then be swept away several tenths of meters within a few seconds to a low-background area where the chemistry is performed.

When using a gas-filled mass separator between the target and the chemistry there is an additional problem because the products have to pass from the low pressure ($\sim$100 Pa) in the mass separator to the high pressure ($\sim$0.2 MPa) in the gas jet transportation system. This is achieved by using a recoil transfer chamber with a thin window that allows the nuclear reaction products to pass from the low pressure to the high pressure area where they attach to the aerosol particles.

17.5. PRODUCT SPECIFICATIONS

In many countries there are one or more suppliers of *radiochemicals*. To locate suppliers, the simplest way is often to browse the internet but it is also possible to contact the nearest nuclear center, as it may be a producer of radionuclides. Other sources of information may be e.g. the national radiation safety organization, or the annual Byers' Guide of common nuclear journals.

The radionuclides come in the form of calibration/reference sources, and as inorganic compounds and labelled organic compounds. They are packed in various ways, most commonly for compounds in a glass ampoule packed in a small aluminium can. To reduce the surface radiation dose rate (cf. Ch. 15) for high intensity radiation sources (usually $>$10 MBq) the can is put in a lead block which is packed in a wooden crate for shipment. Opening of the aluminum can and glass ampoule may require remote control and should be done only by experienced, licensed personnel.

17.5.1. Radiochemical processing

In some cases it may be a long process between the removal of the irradiated target from the irradiation facility and the pure radiochemical substance or compound to be used in some experiment. The separation procedures needed to make a pure radiation source are chemically the same and (usually) independent of the "size" (i.e. its total radioactivity), but the experiments become more cumbersome — and often slower — when the radiation source is so large that considerable shielding is needed to perform the work.

Most chemical techniques can be applied in the purification processing: precipitation, ion exchange, solvent extraction, electrodeposition, electrophoresis, distillation, etc. The basic purposes are to eliminate radioactive contaminants and to avoid diluting the radionuclide by isotopic stable atoms. If the radionuclide or radiochemical is to be used in medical and biological work, the sample may have to be sterile. Some publications about radiochemical separation procedures are given in the literature list, §17.9.